AI2002
Artificial Intelligence
Lecture 21

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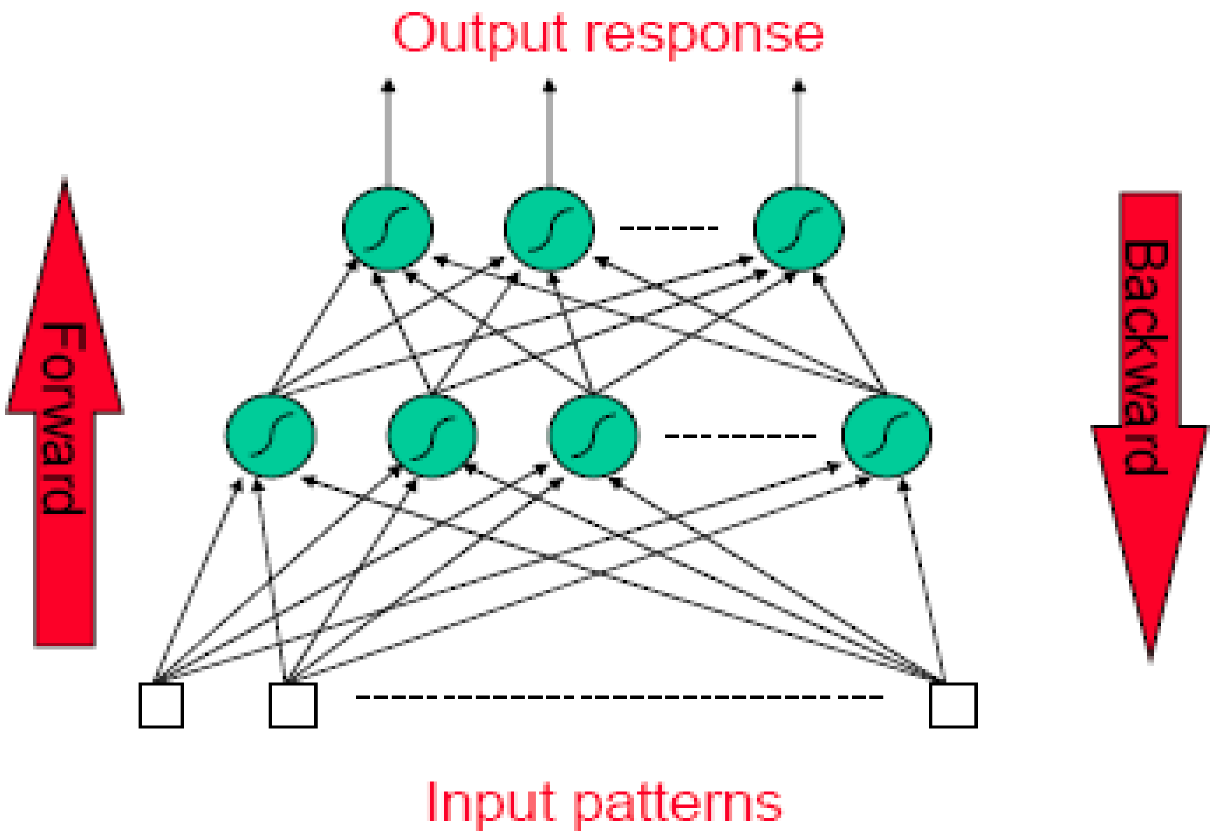
FAST NUCES CFD

Back Propagation Algorithm

The back propagation algorithm having two phase

- **Forward pass phase:**
  - computes 'functional signal', feed forward propagation of input pattern signals through network
- **Backward pass phase:**
  - computes 'error signal', *propagates* the error *backwards* through network starting at output units (where the error is the difference between actual and desired output values)

Back Propagation Algorithm



Back Propagation Algorithm

- The back propagation algorithm learns the weights for a multilayer network,
  - given a network with a fixed set of units and interconnections.

- It **employs gradient descent** to attempt to minimize the squared error between the network output values and the target values for these outputs.

- As we are considering networks with multiple output units, we begin by redefining **E** to sum the errors over all of the network output units.

Back Propagation Algorithm

BACKPROPAGATION($training\_examples, \eta, n_{in}, n_{out}, n_{hidden}$)

Each training example is a pair of the form $\langle \vec{x}, \vec{t} \rangle$, where $\vec{x}$ is the vector of network input values, and $\vec{t}$ is the vector of target network output values.

η is the learning rate (e.g., .05). n_{in} is the number of network inputs, n_{hidden} the number of units in the hidden layer, and n_{out} the number of output units.

The input from unit i into unit j is denoted x_{ji}, and the weight from unit i to unit j is denoted w_{ji}.

- Create a feed-forward network with n_{in} inputs, n_{hidden} hidden units, and n_{out} output units.
- Initialize all network weights to small random numbers (e.g., between $-.05$ and $.05$).
- Until the termination condition is met, Do

Back Propagation Algorithm

For each $\langle \vec{x}, \vec{t} \rangle$ in *training_examples*, Do

Propagate the input forward through the network:

1. Input the instance $\vec{x}$ to the network and compute the output o_u of every unit u in the network.

Propagate the errors backward through the network:

2. For each network output unit k, calculate its error term δ_k

$$\delta_k \leftarrow o_k(1 - o_k)(t_k - o_k)$$

3. For each hidden unit h, calculate its error term δ_h

$$\delta_h \leftarrow o_h(1 - o_h) \sum_{k \in outputs} w_{kh} \delta_k$$

4. Update each network weight w_{ji}

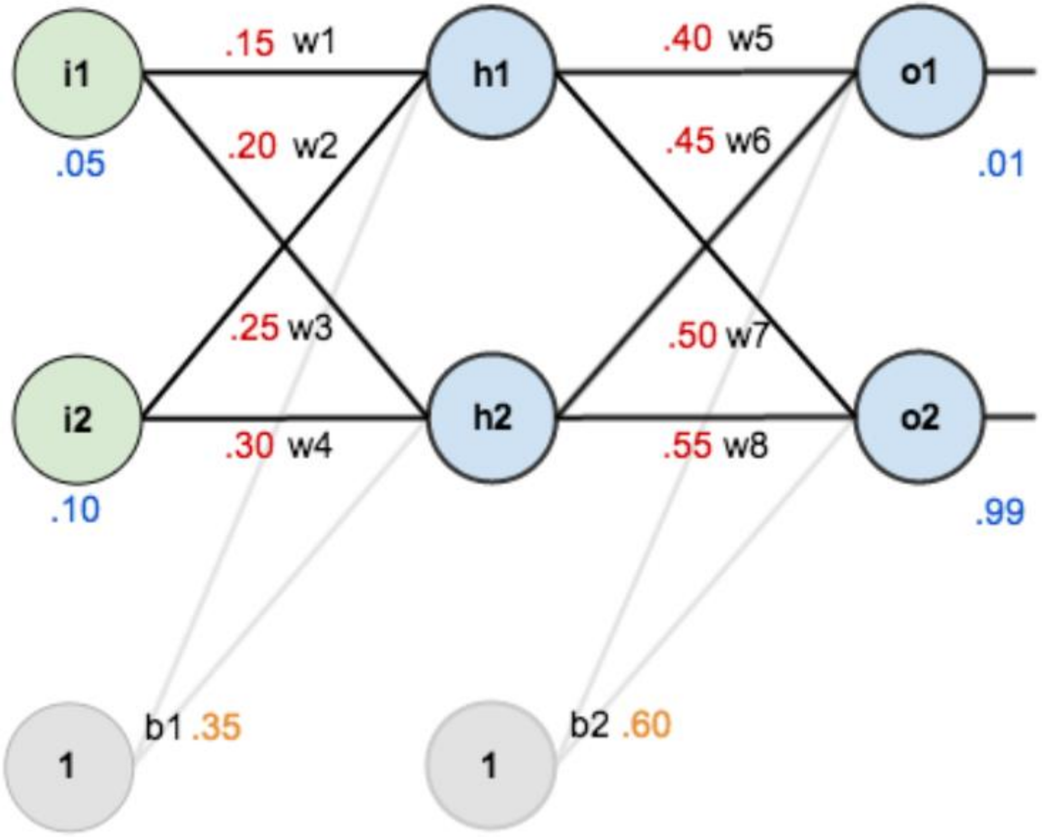
$$w_{ji} \leftarrow w_{ji} + \Delta w_{ji}$$

where

$$\Delta w_{ji} = \eta\, \delta_j\, x_{ji}$$

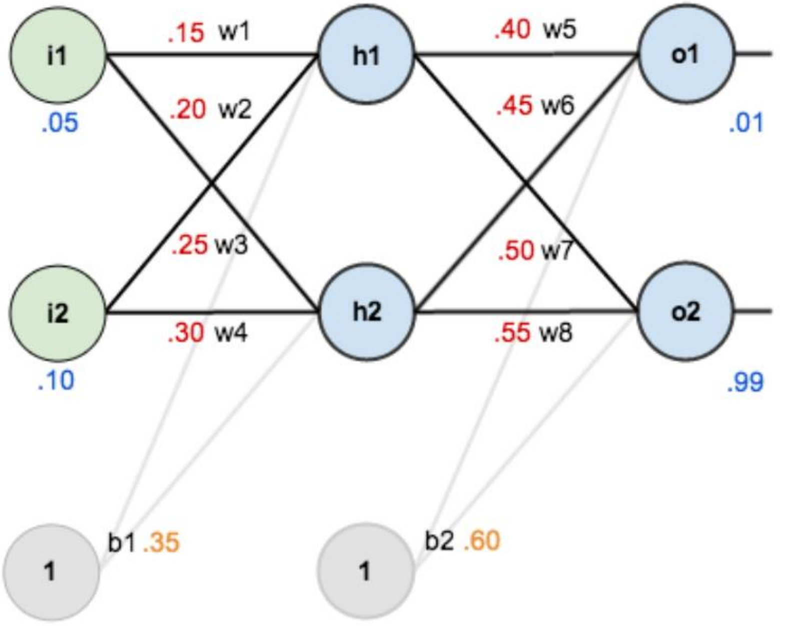
Example:

- Lets consider the following network



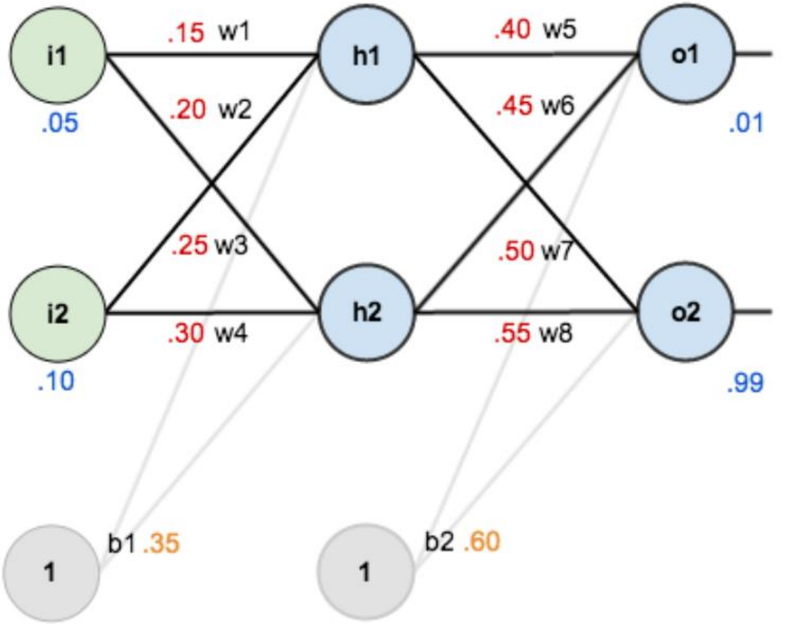
Apply the forward pass phase

- Apply the forward pass phase to compute the output and error
- Output
- $net_{h1} = i1 * w1 + i2 * w2 + (1 * b1)$
- $net_{h1} = (0.15 * 0.05) + (0.2 * 0.1) + (0.35 * 1)$
- $net_{h1} = 0.3775$
- Output of h1
- $out_{h1} = \frac{1}{1+e^{-net}} = \frac{1}{1+e^{-0.3775}} = 0.59326$



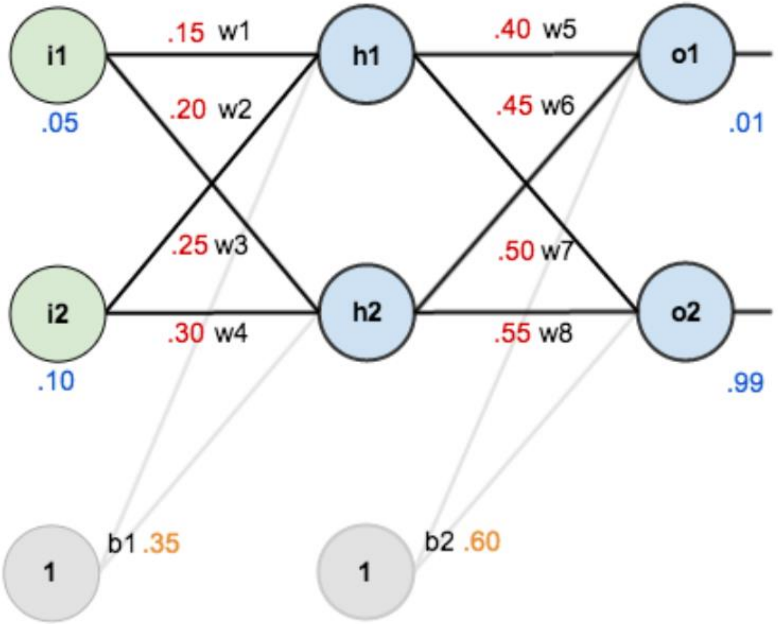
Apply the forward pass phase

- Apply the forward pass phase to compute the output and error
- Output
- $net_{h2} = i1 * w3 + i2 * w4 + (1 * b1)$
- $net_{h2} = (0.15 * 0.25) + (0.2 * 0.3) + (0.35 * 1)$
- $net_{h2} = 0.4475$
- Output of h2
- $out_{h2} = \frac{1}{1+e^{-net}} = \frac{1}{1+e^{-0.4475}} = 0.59688$



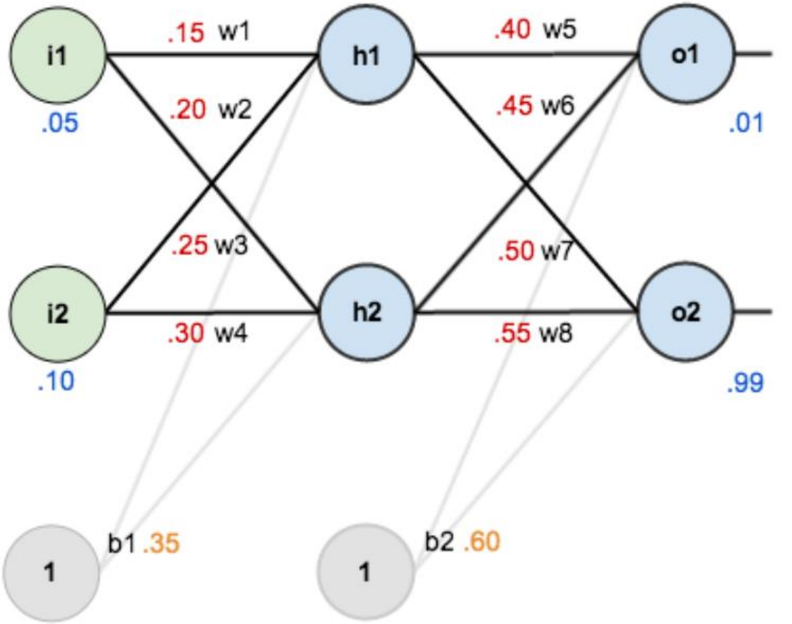
Apply the forward pass phase

- Apply the forward pass phase to compute the output and error
- Output
- $net_{o1} = h1 * w5 + h2 * w7 + (1 * b2)$
- $net_{o1} = (0.59326 * 0.4) + (0.59688 * 0.45) + (0.6 * 1)$
- $net_{o1} = 1.10591$
- Output of h1
- $out_{01} = \frac{1}{1+e^{-net}} = \frac{1}{1+e^{-1.10591}} = 0.75136$



Apply the forward pass phase

- Apply the forward pass phase to compute the output and error
- Output
- $net_{o2} = h1 * w6 + h2 * w8 + (1 * b2)$
- $net_{o2} = (0.59326 * 0.5) + (0.59688 * 0.55) + (0.6 * 1)$
- $net_{o2} = 1.2249$
- Output of h1
- $out_{02} = \frac{1}{1+e^{-net}} = \frac{1}{1+e^{-1.2249}} = 0.77295$



Calculating the total Error

$$E_{total} = \sum \frac{1}{2}(target - output)^2$$

- The Error will computed by the above formula
- Error of o1

$$E_1 = \frac{1}{2}(0.01 - 0.75136)^2 = 0.27481$$

- Error of o2

$$E_2 = \frac{1}{2}(0.99 - 0.77295)^2 = 0.02356$$

Total Error

- Total Error

$$E_{total} = E_1 + E_2$$

$$E_{total} = 0.27481 + 0.02356$$
$$= 0.29837$$

Common Formula

- Calculate the weight for output layer (final)

$$\Delta w_{out} = \eta \; . \; \delta 0_{layername} . \, out_{h1}$$

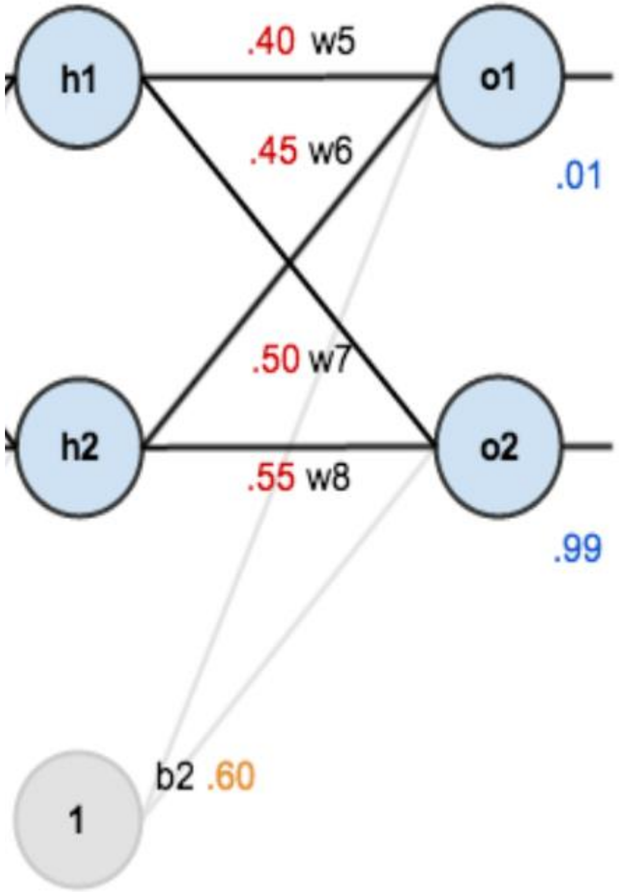
$$\delta_{out} = (t_{01} \text{ - } out_{01}) . \, out_{01}(1 - out_{01})$$

- Calculate the weight for hidden layers

$$\Delta w_{hidden} = \eta \; . \; \delta h_{layername} . \, \delta h_{layername} . \, xinput$$

$$= out_{layername}(1 - out_{layername}) . \sum w_k \delta_k$$

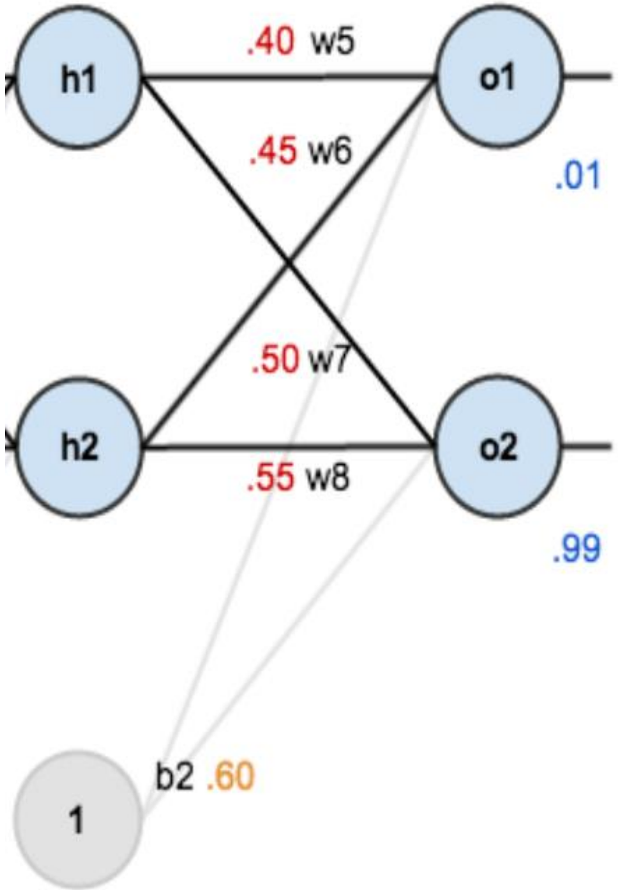
Calculate the weight for output layers



Make the equation

- h1
  - W5 map on o1
    - So for w5 we calculate the error of o1
  - W6 map on o2
    - So for w6 we calculate the error of o2

- H2
  - W7 map on o1
    - So for w7 we calculate the error of o1
  - W8 map on o2
    - So for w8 we calculate the error of o2



Equation to Calculate the error

- Output layer o1

$$\delta_{01} = (t_{01} \text{ - } out_{01}).\, out_{01}(1 - out_{01})$$

- Output layer o2

$$\delta_{02} = (t_{02} \text{ - out}t_{02}).\, out_{02}(1 - out_{02})$$

Equation to Calculate the error

- Output layer o1

$$\delta_{01} = (t_{01} \text{ - } \mathrm{out}_{01}).\, out_{01}(1 - out_{01})$$

- out_{01} = 0.75136
- $t_{01} =$ 0.01
- δ_{01} = (0.01 - 0.75136)(0.75136)(0.24864)
- δ_{01} =-0.1397

Equation to Calculate the error

- Output layer o2

$$\delta_{02} = (t_{02} \text{ - out}_{02}).\, out_{02}(1 - out_{02})$$

- $out_{02} = 0.77295$
- t_{02} = 0.99
- $\delta_{02} = (0.99–0.77295)·0.77295·(1–0.77295)$
- $\delta_{02} = 0.0377$

Equation for weights of hidden layer

- Output layer o1

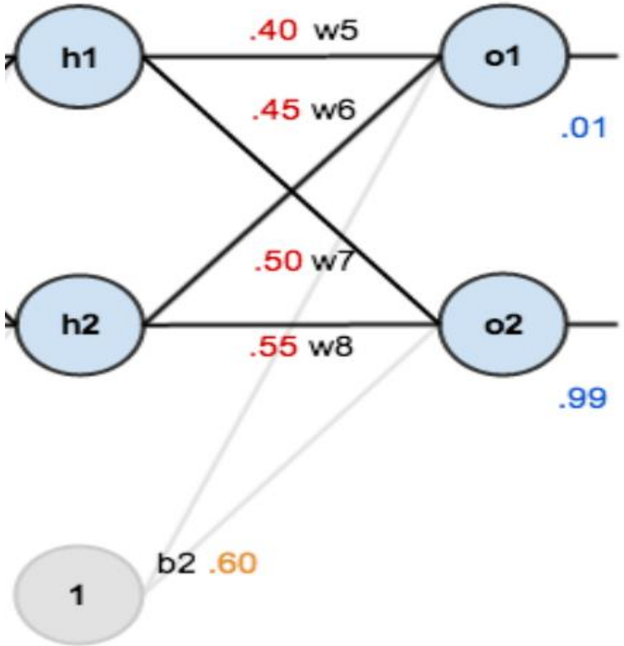
$$\delta_{01} = (t_{01} \text{ - out}_{01}).out_{01}(1 - out_{01})$$

- $\Delta w_5 = w_5 + \Delta w_5$
- $\Delta w_5 = \eta . \delta 0_1 . out_{h1}$

- $\Delta w_6 = w_6 + \Delta w_6$
- $\Delta w_6 = \eta . \delta 0_2 . out_{h1}$

- $\Delta w_7 = w_7 + \Delta w_7$
- $\Delta w_7 = \eta . \delta 0_1 . out_{h2}$

- $\Delta w_8 = w_8 + \Delta w_8$
- $\Delta w_8 = \eta . \delta 0_2 . out_{h2}$



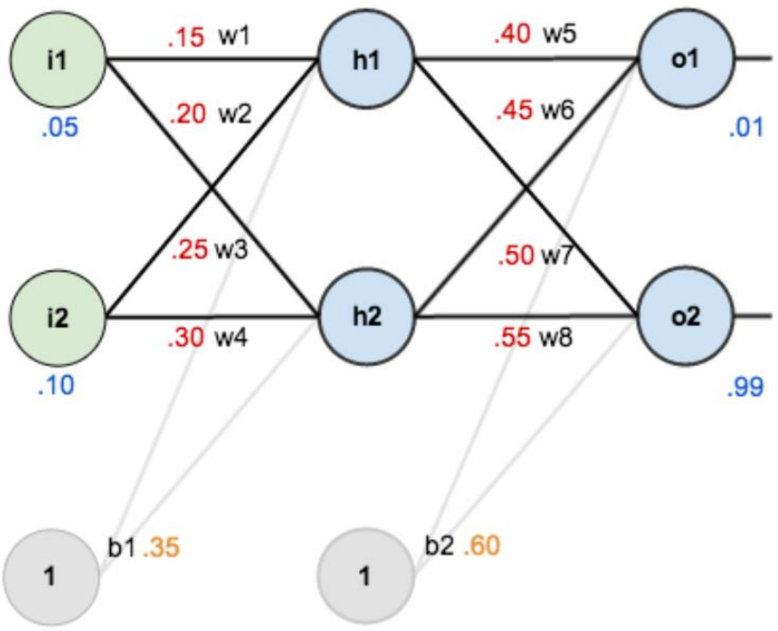
Calculate the weight for hidden layer

- Output layer h1

$$\delta_{h1} = (1 - out_{h1}).\, out_{h1}(w_5\delta_{01} + w_6\delta_{02})$$

- Output layer h2

$$\delta_{h2} = (1 - out_{h2}).\, out_{h2}(w_7\delta_{01} + w_8\delta_{02})$$



Calculate the weight for hidden layer

- Output layer h1

$$\delta_{h1} = (1 - out_{h1}).\, out_{h1}(w_5\delta_{01} + w_6\delta_{02})$$

$$out_{h1} = 0.5922$$

$$\delta_{01} = 0.1385$$

$$\delta_{02} = -0.0381$$

$$w_5 = 0.4$$

$$w_6 = 0.5$$

$$=(0.1385 \cdot 0.40 + (-0.0381) \cdot 0.50) \cdot 0.5933 \cdot (1 - 0.5933)$$

$$=0.0052$$

- Output layer h2

$$\delta_{h2} = (1 - out_{h2}).\ out_{h2}(w_6\delta_{01} + w_8\delta_{02})$$

$$out_{h2} = 0.5968$$

$$\delta_{01} = 0.1385$$

$$\delta_{02} = -0.0381$$

$$w_7 = 0.45$$

$$w_8 = 0.55$$

$$=(0.1385 \cdot 0.45 + (-0.0381) \cdot 0.55) \cdot 0.5969 \cdot (1 - 0.5969)$$

$$= 0.0059$$

Equations to update the weight of hidden layer

- Output layer h1

$$\delta_{h1} = (1 - out_{h1}).\, out_{h1}(w_5\delta_{01} + w_7\delta_{02})$$

- Output layer h2

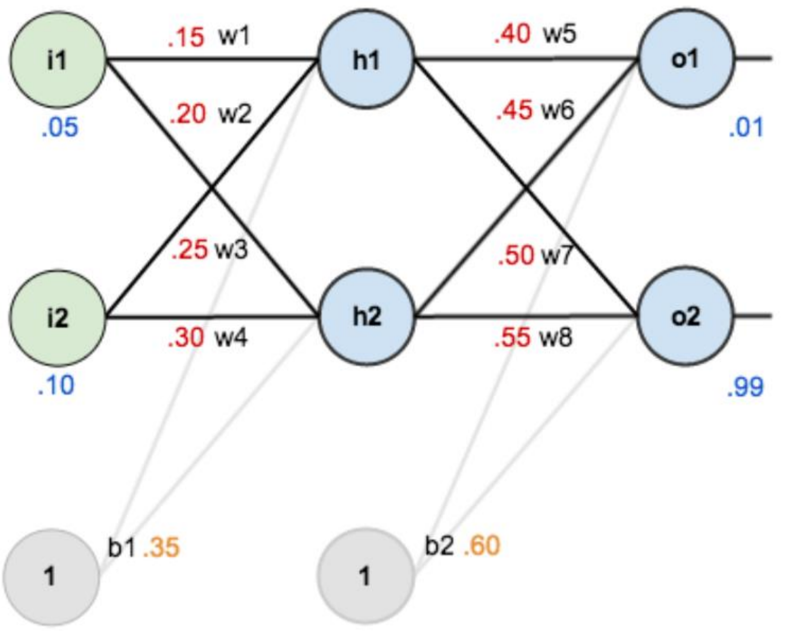
$$\delta_{h2} = (1 - out_{h2}).\, out_{h2}(w_6\delta_{01} + w_8\delta_{02})$$

- $\Delta w_1 = w_1 + \Delta w_1$
- $\Delta w_1 = \eta.\, \delta_{h1}.\, i_1$

- $\Delta w_2 = w_2 + \Delta w_2$
- $\Delta w_2 = \eta.\, \delta_{h2}.\, i_1$

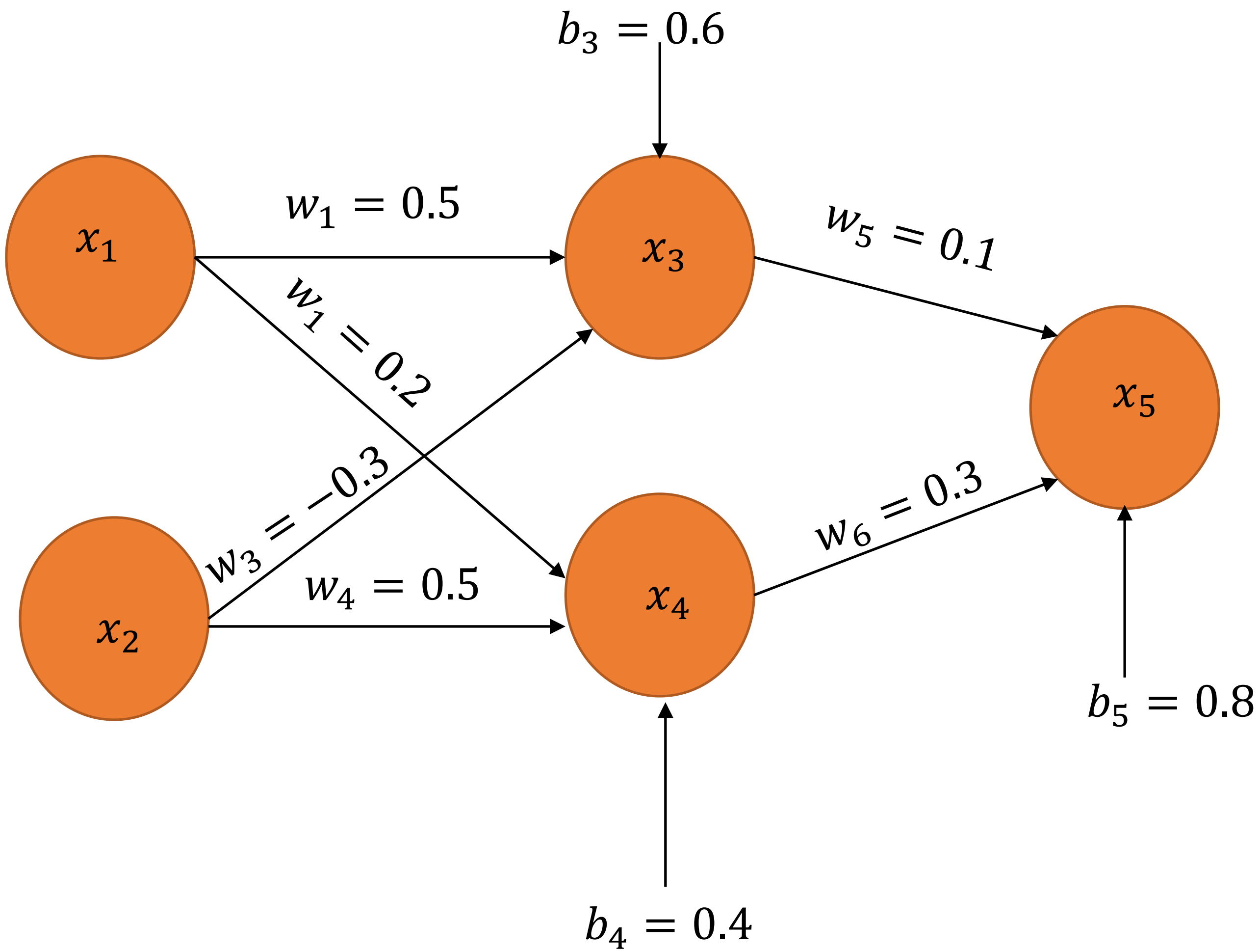
- $\Delta w_3 = w_3 + \Delta w_3$
- $\Delta w_3 = \eta.\, \delta_{h1}.\, i_2$

- $\Delta w_4 = w_4 + \Delta w_4$
- $\Delta w_4 = \eta.\, \delta_{h2}.\, i_2$



Example 2:

x_1	x_1	t_{output}
1	1	0



Apply the feed Forward Pass

- $O.x_3 = (x_1.w_1).(x_2.w_2).(b_3)$
  - $O.x_3 = (1 \times 0.5).(1 \times -0.3)(0.6)$
  - O.x_3 = 0.8
  - $x_3 = \frac{1}{1+e^{-O(x_3)}} = \frac{1}{1+e^{-0.8}}$ = 0.69

- $O.x_4 = (x_1.w_3).(x_2.w_4).(b_4)$
  - $O.x_4 = (1 \times 0.2).(1 \times 0.5)(0.4)$
  - O.x_4 = 0.3
  - $x_4 = \frac{1}{1+e^{-O(x_4)}} = \frac{1}{1+e^{0.3}}$ = 0.5744

Apply the feed Forward Pass

- $O.x_5 = (x_3.w_5).(x_4.w_6).(b_5)$
  - $O.x_5 = (0.69 \times 0.1).(0.5744 \times 0.3)(0.8)$
  - O.x_5 = 1.0413
  - $x_5 = \frac{1}{1+e^{-O(x_5)}} = \frac{1}{1+e^{1.0413}}$ = 0.7391

Backward Propagation Equation

- Calculate the weight for output layer (final)

$$\Delta w_{out} = \eta \, . \, \delta 0_{layername} . \, out_{h1}$$

$$\delta_{out} = (t_{01} \text{ - } out_{01}). \, out_{01}(1 - out_{01})$$

- Calculate the weight for hidden layers

$$\Delta w_{hidden} = \eta \, . \, \delta h_{layername} . \, xinput$$

$$= out_{layername}(1 - out_{layername}). \sum w_k \delta_k$$

Backward Propagation Equation

- Output layer o1

$$\delta_{x5} = (t_{x5} \text{ - out}_{x5}).\,out_{x5}(1 - out_{x5})$$
$$\delta_{x5} = (0 - 0.7391)(0.7391)(1 - 0.7391)$$
$$\delta_{x5} = -0.1425$$

- $\Delta w_5 = w_5 + \Delta w_5$
- $\Delta w_5 = w_5 + \eta.\ \delta_{x5}.\,x_3$ = 0.1 + ((0.5)(-0.1425)(0.69))
- w_5 = 0.0508

- $\Delta w_6 = w_6 + \Delta w_6$
- $\Delta w_6 = w_6 + \eta.\delta_{x5}.\,x_4$ =0.6 + ((0.3)(-0.1425)(0.5744))
- w_6 =0.2591

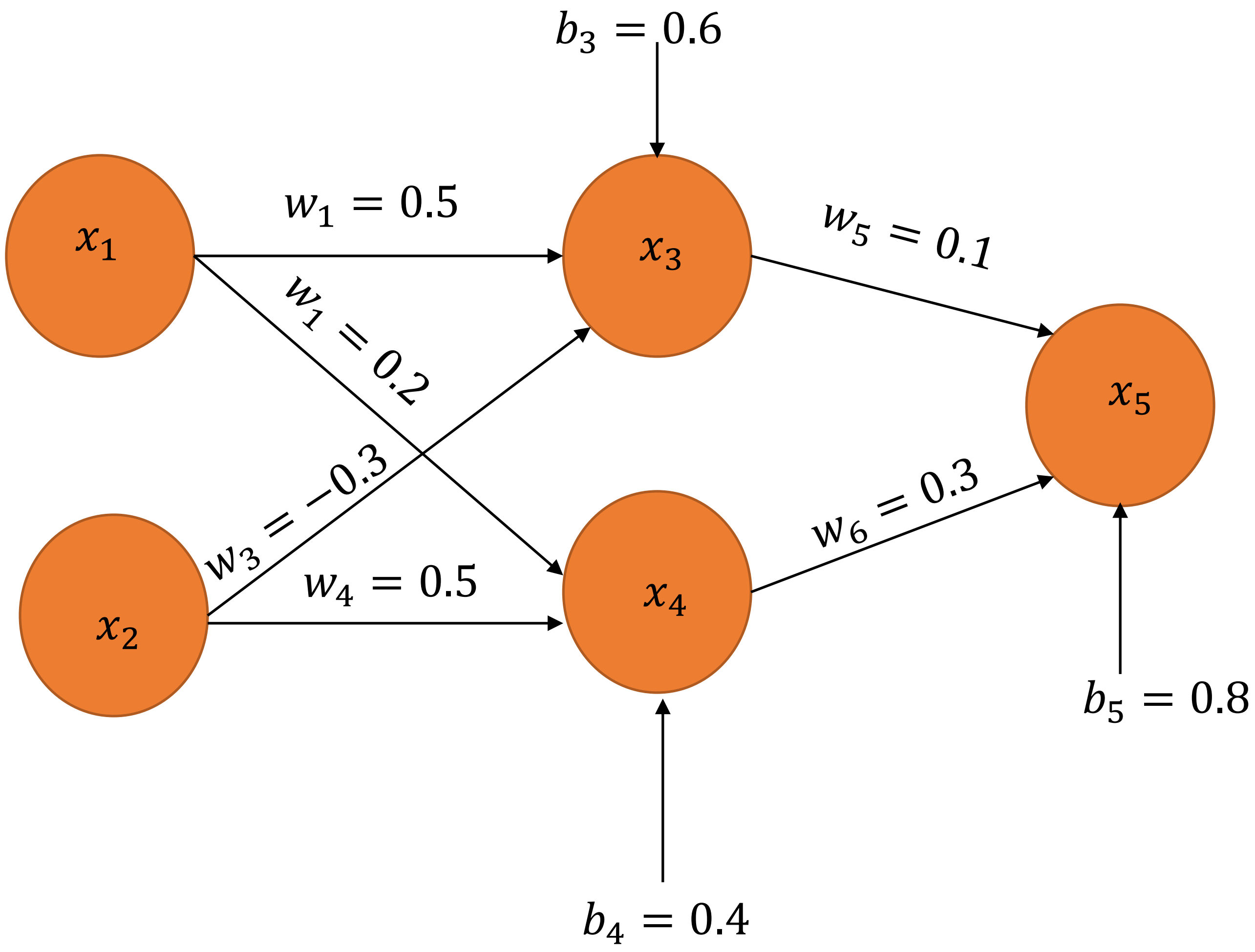
Hidden Layers

- $\delta x_3 = x_3(1 - x_3).\sum w_k \delta_k$
- $\delta x_3 = x_3(1 - x_3).(w_5\ \delta_{x5})$
- δx_3 = (0.69)(1 - 0.69)(0.1*-0.1425)
- δx_3 = -0.030481

- $\delta x_4 = x_4(1 - x_4).\sum w_k \delta_k$
- $\delta x_4 = x_4(1 - x_4).(w_6\ \delta_{x5})$
- δx_4 = (0.5744)(1 - 0.5744)(0.1*-0.1425)
- δx_4 = -0.010451

Example 2:

x_1	x_1	t_{output}
1	1	0



Weights Update

- Output layer hidden layers

δx_3 = -0.030481

δx_4 = -0.010451

- $\Delta w_1 = w_1 + \Delta w_1$
- $\Delta w_1 = w_1 + \eta.\ \delta_{x3}.\ x_1$

- $\Delta w_2 = w_2 + \Delta w_2$
- $\Delta w_2 = w_2 + \eta.\delta_{x4}.\ x_2$

- $\Delta w_3 = w_3 + \Delta w_3$
- $\Delta w_3 = w_3 + \eta.\ \delta_{x3}.\ x_3$

- $\Delta w_4 = w_4 + \Delta w_4$
- $\Delta w_4 = w_4 + \eta.\ \delta_{x4}.\ x_4$